

CPRE 4880 - Section A1

MP - 1

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Dr. Phillip Jones

In your write-up, describe your experience in practicing controlling the quad.

In our experience trying to get enough thrust to overcome the ground effect was the main difficulty for trying to maintain a nice level flight, the control over the yaw and pitch was a learning curve but being very careful and slow with our controller movements we were able to get a smooth control over the drone.

In your write-up, describe the PPM signals. What do each of the channels correspond to, and what are their minimum and maximum ranges? What is the total length of the individual PPM frames, and what is the minimum length of the idle pulse?

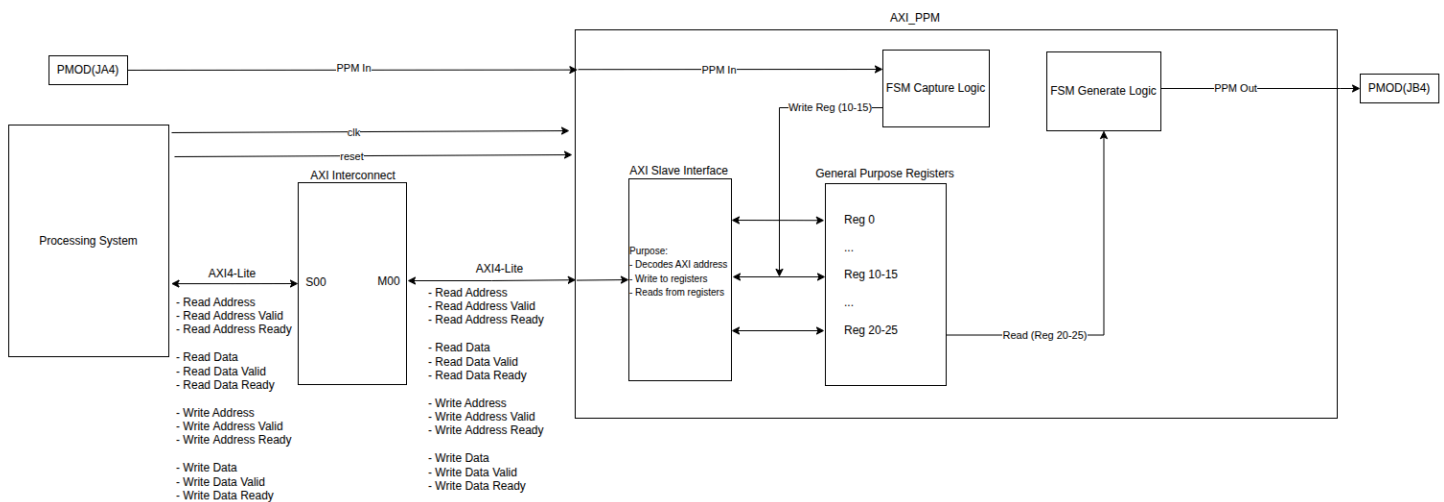
Min range : 700 μ s Max Range : 1.5m

Channels	Relation	Min Range	Max Range
1	Right Stick Side to Side	786.4 μ s (left)	1.55 ms (right)
2	Right Stick up and down	763.6 μ s (down)	1.436 ms (up)
3	Left Stick up and down	706.6 μ s (down)	1.55 ms (up)
4	Left Stick Side to Side	740.8 μ s (left)	1.527 ms(right)
5	Right knob	604 μ s (left)	1.63 ms (right)
6	Left knob	604 μ s(left)	1.63 (right)
Idle		398.9 μ s	410.3 μ s
PPM Frames		19.95 ms	20.05 ms

Based on the ZedBoard documentation and your oscilloscope measurement of the trainer port, what concerns do you have about making this connection?

The biggest concern is the voltage ranges on each device, we see that on the trainerport its running on 5V while the Zedboard can only handle 3.3V on the PMOD interface, thus we are going to need to insert a level shifter so that we can properly send our PPM signal into the PMOD without damaging its hardware.

1) In your write-up, provide a structural diagram of the axi_ppm design, from the top-level AMBA AXI interface down to where your user logic will reside.



2) How does an address on the AMBA bus generate a read or write enable signal for the slave registers in your design?

When the processing system performs a read or write instruction, it places an address on the AXI bus. The AXI Interconnect compares the upper address bits against its configured address map and routes to our slave `axi_ppm` if the address falls within the assigned range.

Once the transaction reaches the AXI slave interface inside the `axi_ppm`, the lower address bits (the offset) are decoded. During a write transaction, when the AXI handshake signals indicate a valid transfer (`AWVALID/WVALID` and `READY` are asserted), the slave interface asserts an internal write enable signal (`slv_reg_wren`). The decoded address bits are used in a case statement to select which slave register (`slv_regX`) is updated with the write data.

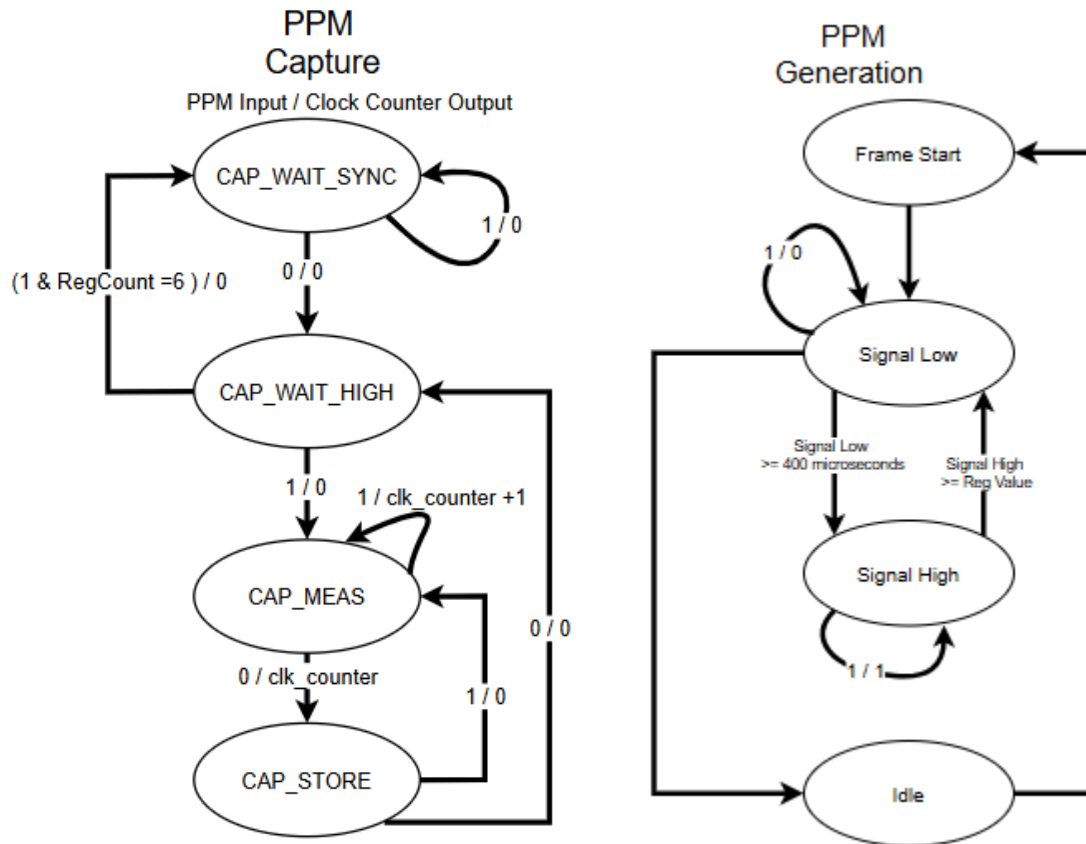
For a read transaction, the decoded address selects the appropriate slave register, and its value is multiplexed onto the AXI read database which is returned to the Processing System.

3) How will your PPM state machine get access to the IP core's Memory-Mapped registers?

The PPM state machines will get access to the memory mapped registers because those registers are internal signals inside the same IP core. In other words our VHDL code will already contain the slave registers declared as signals inside our IP Core.

For more detail, `slv_reg0` and `slv_reg1` are only going to be written by software and can not have access by the PPM state machines. `slv_reg10-15` can only be accessed by the PPM state machines and not by the processing system. `slv_reg20-25` can only be written by the processing system and not by the PPM state machines.

Starting from the generic FSMs in your HW-1 write-up, update the User Logic module such that the design can appropriately capture and generate PPM values for the Hobby King 6ch transmitter as described above.



Bonus Credit:

None :(

Student Effort

The following represents the percentage of effort from each team member for MP-0:

Cameron Gilbertson:	25%
Matthew Smith:	25%
Jessica Reiff:	25%
Collin Van Weelden:	25%

